



Got your back!

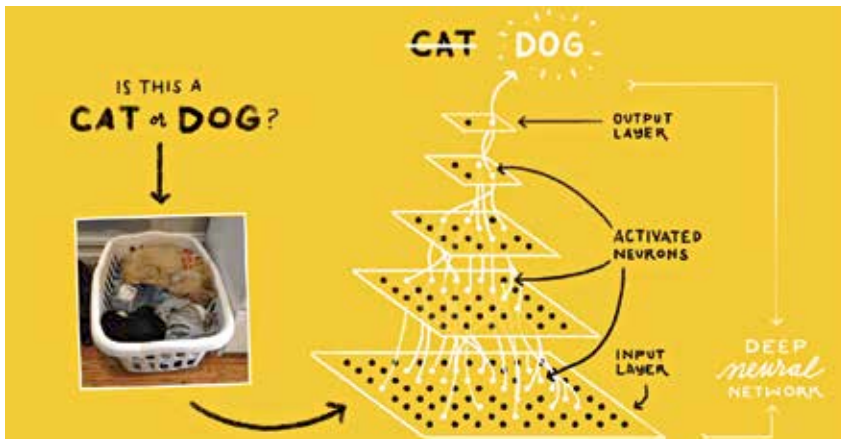
OmniWear Haptics' Arc can warn you when enemies sneak up behind you during your FPS gameplay. <http://digit.in/OWrHArc>



1050 and 1050Ti are here

Compatible with prebuild PCs, NVIDIA's latest 1050 and 1050 Ti graphic cards launched for Rs. 10,000 and 12,500 respectively. <http://digit.in/1050Lnchd>

Tomorrow's Tech



Google's TensorFlow framework's open source, designed for machine learning apps

Beyond the marketing hype

Of course, TPUs are just plain and simple ASICs, at the end of the day, whatever Google may call them. There's a lot of merit in designing and manufacturing application-specific hardware as much as possible, to drive up efficiency of the system overall. Bitcoin mining is a very specific activity, requiring the system to do it ever more efficiently, for instance. The better a circuit's design to mine bitcoin, obviously the better it will perform.

While Google hasn't really designed and created a piece of silicon from scratch before – at least from what's public knowledge – it has a history of tinkering with commercially purchased, off the shelf hardware to best suit its work environment. Remember back in 2009, when news first broke about a custom-made Google server with a unique onboard UPS battery unit fitted onto the board itself to ensure power efficiency and redundancy? The company has understood long ago that the success of its critical software-driven infrastructure depends closely on efficient hardware systems powering it all.

Let's get one thing straight – Google's not suddenly becoming a silicon provider like NVIDIA or Intel. That's not its traditional strength. Which means it's not going to pose an immediate threat to companies that fabricate and manufacture silicon chips. But – and this is an important but – Google has demonstrated very clearly that through TPUs it plans to enable the Google Cloud

Platform and TensorFlow ecosystem to race far ahead of its competitors.

In fact, analysts believe that NVIDIA with its Pascal-based GPU is probably the best equipped hardware platform for Deep Learning AI. That's right, not just used for calculating polygons and graphics, NVIDIA Pascal-based GPU stacks are great at numerical floating point calculations at a record pace. In fact, NVIDIA's P100 is considered as a behemoth of the machine learning / AI industry, primarily used for training or increasing a system's learning capability – a far bigger challenge in machine learning parlance than 'inferencing' or predicting an outcome. Google hasn't confirmed it yet, but word on the street is that TPUs are most likely being used for inferencing purpose in the company's machine learning endeavours.

Healthy competition

Google obviously has a lot to share about TPUs, and our guess is that it's deliberately holding most of its secret hardware cards close to its chest – it has



The NVIDIA P100 chip

every right to do so. What's happening as a result is that the AI / deep learning / machine learning industry as a whole is trying to ramp up their operations – both on the hardware (Intel, NVIDIA, IBM, etc) and cloud-software side (Microsoft Azure, Amazon Web Services and EC2, etc) of things. This healthy competition between these industry heavyweights will ensure the pace of innovation doesn't lose steam, that in fact it picks up by several notches, and that's a good sign for the future of machine learning and evolution of AI.

TPUs are a sign of things to come. It's a signal that the machine learning market at large is moving from an exploratory phase – where it was okay for companies to throw money at unearthing effective, accurate solutions – to a commercialization phase – where it's now imperative for companies to find cost-effective, ever-efficient solutions on top of being



FPGAs can give TPUs stiff competition

accurate. This means startups spawning in the machine learning / neural networks space now have options to choose from on what they want to do. The potential of FPGAs or Field Programmable Gate Array – which allows customers to reconfigure the chip after it's manufactured – also becomes extremely important, giving stalwarts like Intel and Microsoft something to pit against Google's TPU prowess.

In the race to develop the best AI, on the foundation of highly-efficient machine learning systems, power usually resides with those who have the most data. And if there's one thing about Google that we know for sure, something that even Facebook will begrudgingly agree to, is that they have a LOT of data. **T**